

1. NumPy Basics (1–20)

1. What is NumPy?
2. Why is NumPy faster than Python lists?
3. What is an ndarray?
4. How do you create a NumPy array?
5. Difference between a Python list and a NumPy array?
6. What are the advantages of NumPy?
7. How do you check the version of NumPy?
8. How do you find the shape of an array?
9. What is the difference between `shape`, `size`, and `ndim`?
10. How do you find the data type of an array?
11. How do you change the data type of an array?
12. What are NumPy scalar types?
13. How do you create an array of zeros?
14. How do you create an array of ones?
15. How do you create an identity matrix?
16. Difference between `zeros()` and `empty()`?
17. What does `arange()` do?
18. Difference between `arange()` and `linspace()`?
19. How do you create random arrays?
20. What is the default data type in NumPy?

2. Indexing & Slicing (21–35)

21. How do you access a single element?
22. How do you slice a NumPy array?
23. Difference between slicing in lists and NumPy?
24. What is negative indexing?
25. How do you access rows and columns?
26. How do you reverse an array?
27. What is fancy indexing?
28. What is Boolean indexing?
29. How do you filter an array?
30. How do you replace values conditionally?
31. How do you find elements greater than a value?
32. How do you extract every second element?
33. How do you select multiple rows?
34. Difference between copy and view?
35. How can you check whether an array owns its data?

3. Array Manipulation (36–50)

- 36. How do you reshape an array?
 - 37. Difference between `reshape()` and `resize()`?
 - 38. What happens if reshape dimensions don't match?
 - 39. What does `flatten()` do?
 - 40. Difference between `flatten()` and `ravel()`?
 - 41. How do you transpose an array?
 - 42. Difference between transpose and reshape?
 - 43. How do you concatenate arrays?
 - 44. Difference between `concatenate()`, `stack()`, `vstack()`, and `hstack()`?
 - 45. How do you split arrays?
 - 46. What is `column_stack()`?
 - 47. What is `row_stack()`?
 - 48. How do you repeat elements?
 - 49. How do you tile arrays?
 - 50. How do you insert or delete elements?
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4. Mathematical Operations (51–65)

- 51. How do arithmetic operations work in NumPy?
 - 52. Explain broadcasting.
 - 53. Broadcasting rules?
 - 54. What are universal functions (ufuncs)?
 - 55. Difference between `np.sum()` and Python `sum()`?
 - 56. Difference between `mean()` and `average()`?
 - 57. How do you calculate variance?
 - 58. How do you calculate standard deviation?
 - 59. How do you find minimum and maximum values?
 - 60. Difference between `argmax()` and `max()`?
 - 61. How do you calculate cumulative sums?
 - 62. How do you compute cumulative products?
 - 63. How do you round numbers?
 - 64. Difference between `floor()`, `ceil()`, and `round()`?
 - 65. How do you compute logarithms and exponentials?
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5. Matrix Operations (66–75)

- 66. Difference between arrays and matrices?
- 67. How do you perform matrix multiplication?
- 68. Difference between `*` and `@` operators?
- 69. What does `np.dot()` do?
- 70. Difference between `dot()` and `matmul()`?
- 71. How do you compute matrix inverse?
- 72. How do you calculate determinant?
- 73. How do you find eigenvalues?

- 74. How do you solve linear equations?
- 75. What is Singular Value Decomposition (SVD)?

6. Random Module (76–82)

- 76. How do you generate random integers?
- 77. Difference between `rand()`, `randn()`, and `random()`?
- 78. How do you set a random seed?
- 79. Why is reproducibility important?
- 80. How do you shuffle an array?
- 81. Difference between `shuffle()` and `permutation()`?
- 82. How do you sample without replacement?

7. Performance & Optimization (83–90)

- 83. Why is NumPy faster?
- 84. What is vectorization?
- 85. Why should Python loops be avoided in NumPy?
- 86. What is broadcasting optimization?
- 87. What are memory views?
- 88. Difference between contiguous and non-contiguous arrays?
- 89. What is memory alignment?
- 90. How do you measure NumPy performance?

8. Advanced NumPy (91–100)

- 91. What are structured arrays?
- 92. What are masked arrays?
- 93. What is memory mapping (`mmap`)?
- 94. What is `einsum()`?
- 95. Explain strides.
- 96. What is stride trick?
- 97. Difference between shallow copy and deep copy?
- 98. How do NumPy arrays interact with Pandas?
- 99. How does NumPy integrate with machine learning libraries?
- 100. What are the limitations of NumPy?

Coding Interview Questions

1. Reverse a NumPy array.
2. Find unique elements.
3. Remove duplicate values.
4. Find the second largest element.
5. Rotate an array.
6. Sort rows independently.
7. Find missing values (NaN).
8. Replace NaN with column means.
9. Normalize an array.
10. Standardize an array.
11. Compute Euclidean distance.
12. Compute cosine similarity.
13. One-hot encode labels.
14. Create a checkerboard matrix.
15. Find all prime numbers in an array.
16. Count occurrences of each element.
17. Find the most frequent value.
18. Perform min-max scaling.
19. Compute moving averages.
20. Implement softmax using NumPy.